

LAB REPORT

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E-Mail Forensics

Name : Sarthak Das

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For the class of Computer Forensic, taught by

Drabanti Boral

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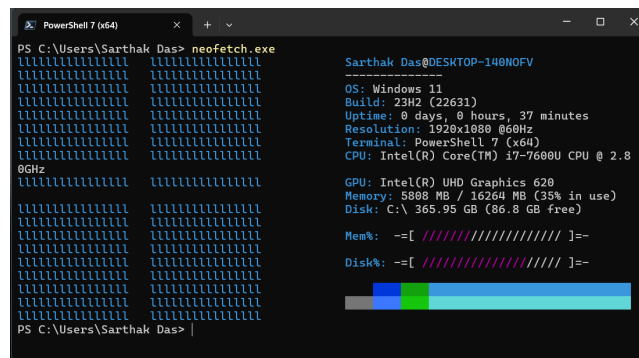
1 Introduction

1.1 Objectives

This report is a documentation of my lab work on e-mail forensics. The primary objective of this lab is to analyze suspicious email messages using SysTools EML Viewer and MxToolbox.

1.2 System Information

For the purposes of this lab, a Lenovo ThinkPad X270 was used, having the following system specifications: Intel Core i7-7600U CPU @ 2.80GHz with Intel UHD Graphics 620, 16GB RAM, 1TB SSD running operating system Windows 11 23H2 (Build 22631).



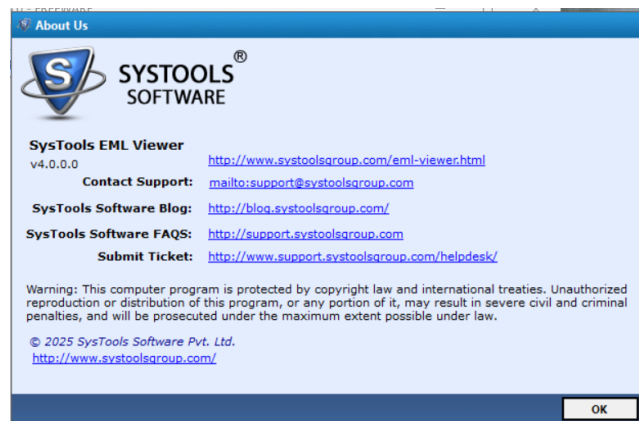
```
PS C:\Users\Sarthak Das> neofetch.exe

Sarhak Das@DESKTOP-148NOFV
OS: Windows 11
Build: 23H2 (22631)
Uptime: 0 days, 0 hours, 37 minutes
Resolution: 1920x1080 @60Hz
Terminal: PowerShell 7 (x64)
CPU: Intel(R) Core(TM) i7-7600U CPU @ 2.8
GHz
GPU: Intel(R) UHD Graphics 620
Memory: 5808 MB / 16264 MB (35% in use)
Disk: C:\ 365.95 GB (86.8 GB free)

Mem%:  --[  //////////////// ]--
Disk%:  --[  //////////////// ]--
```

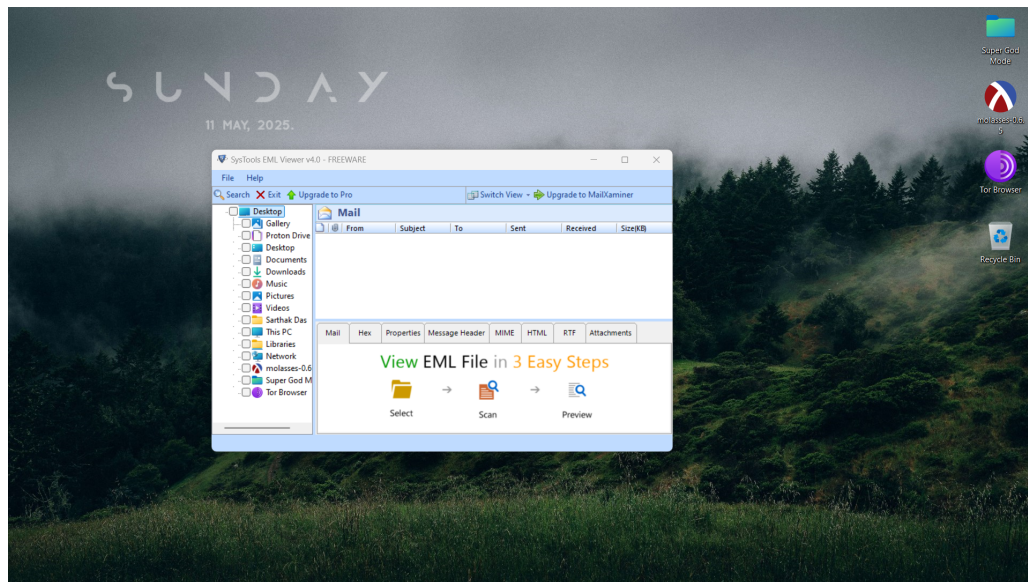
1.3 Software Used

The software used for this practical are SysTools EML Viewer v4.0.0 and the [MxToolbox website](https://www.mxtoolbox.com).

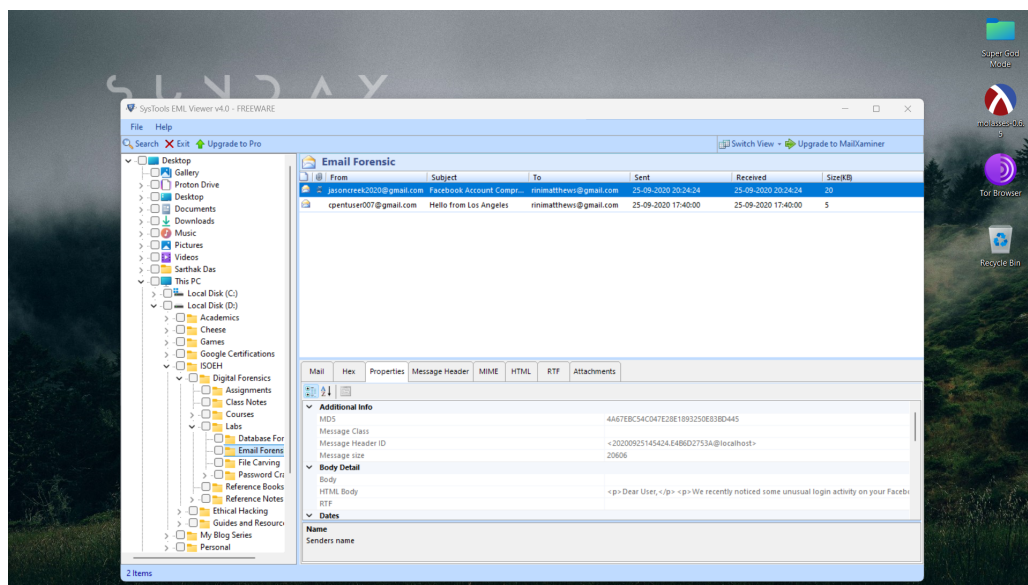


2 Objective: E-mail Browsing

We begin by opening SysTools EML Viewer, and selecting the folder in the file browser where the EML files are located. For the purposes of this lab, we were provided with two EML files – *Normal Email Message.eml* and *Suspicious Email Message.eml*.



Once opened, we can read the e-mails and inspect the e-mail metadata to flag suspicious activity.



3 Objective: Analysis

The suspicious e-mail message reads as follows:

“Dear User,

We recently noticed some unusual login activity on your Facebook account. Therefore, for security reasons, we have disabled your account. To recovery your Facebook account, you ned to reset your password. To reset your Facebook password, either click the link given below, or read the document attached to understand how to reset it.

https://www.facebook.com/help/update_password”

The message is from the GMail domain instead of Facebook. In fact, the GMail domain is spoofed – the domain from which the e-mail has actually been send is detected to be <http://93.99.104.21>. Moreover, the link to Facebook password update actually re-directs to <http://10.0.0.23>. This seems suspicious right off the bat, so we need to investigate further using MxToolbox.

Upon investigating with MxToolbox, we notice that this domain is indeed in the blacklist, implying that this e-mail is a potential spam, hence should be flagged as such.

SuperTool Beta

93.99.104.21 Blacklist Check

blacklist:93.99.104.21 Monitor This Solve Email Delivery Problems

We notice you are on a blacklist. Click here for some suggestions

Checking 93.99.104.21 against 61 known blacklists... Listed 2 times with 1 timeouts

Blacklist	Reason	TTL	Response time		
LISTED	RATS Spam	93.99.104.21 was listed	Detail	2100	1 ignore
LISTED	ZayBL	93.99.104.21 was listed	Detail	2100	324 ignore
OK	OSPAM				91
OK	OSPAM RBL				91
OK	Abuse Mail Intelligence Blacklist				2
OK	Abuse Mail Intelligence Domain Blacklist				2
OK	Abuse Mail Intelligence Exploit list				2
OK	Anomali DNSBL				192
OK	BACKSCATTERER				1
OK	BARRACUDA				24
OK	BLOCKLISTDE				1
OK	CALVERT				20
OK	CYMRU BLACKLIST				41

Free MxToolbox Account
Get one (1) Free Monitor to alert you to Email Delivery Issues

Delivery Center
Real-time insight into the Email Deliverability

Inbox Placement
Know if your campaigns will make the inbox

Recipient Complaints
Get feedback on how recipients perceive your email: complaints, unsubscribes, failures, and more...

Adaptive Blacklist Monitoring
Real-time monitoring of all your domain's sending IPs

Mailflow Monitoring
Round-the-clock monitoring of your email delivery

SPF Flattening
Prevent soft delivery failures and easily manage complex SPF configurations